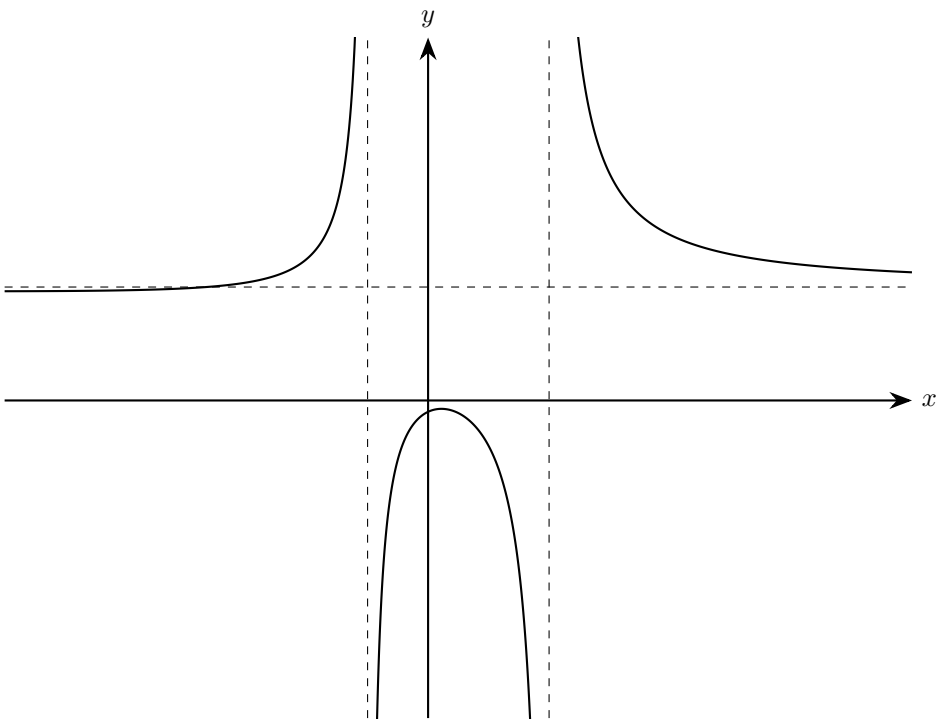
CORE PURE / MISCELLANEOUS PURE

Rational Functions

Worked solutions 14 questions · 177 marks

Questions

Question 1	*8 marks*	2
Question 2	*10 marks*	4
Question 3	*9 marks*	6
Question 4	*9 marks*	8
Question 5	*12 marks*	10
Question 6	*14 marks*	12
Question 7	*14 marks*	16
Question 8	*15 marks*	19
Question 9	*12 marks*	23
Question 10	*15 marks*	26
Question 11	*15 marks*	29
Question 12	*16 marks*	33
Question 13	*14 marks*	37
Question 14	*14 marks*	41



1. The diagram shows the curve C with equation

$$y = \frac{5x^2 + ax - b}{x^2 + bx + a}$$

where a and b are integers.

The vertical asymptotes of C are $x = -1$ and $x = 2$.

(a) State the equation of the asymptote of C that is parallel to the x-axis. **[1]**

(b) Use the vertical asymptotes to find the value of a and the value of b. **[2]**

(c) Hence, or otherwise, find the coordinates of the point where C meets the y-axis. **[1]**

(d) Without using calculus, show that the line $y = 4$ does not intersect C. **[4]**

Solution

(a) The numerator and denominator have the same degree, so $y \to$ the ratio of the leading coefficients as $x \to \pm\infty$. The horizontal asymptote is

$$y = \frac{5}{1} = 5$$

(b) Vertical asymptotes occur where the denominator is zero.

So the quadratic

$$x^2 + bx + a$$

has roots $x = -1$ and $x = 2$.

Hence

$$x^2 + bx + a = (x+1)(x-2)$$

Expanding:

$$(x+1)(x-2) = x^2 - x - 2$$

Comparing with $x^2 + bx + a$, we get

$$b = -1, \qquad a = -2$$

(c) The curve meets the y-axis when $x = 0$.

Using $a = -2$ and $b = -1$:

$$y = \frac{5(0)^2 + a(0) - b}{(0)^2 + b(0) + a} = \frac{-b}{a} = \frac{-(-1)}{-2} = \frac{1}{-2} = -\frac{1}{2}$$

So the point is

$$\left(0, -\frac{1}{2}\right)$$

(d) Using $a = -2$ and $b = -1$, the curve is

$$y = \frac{5x^2 - 2x + 1}{x^2 - x - 2}$$

To find any intersection with the line $y = 4$, solve

$$\frac{5x^2 - 2x + 1}{x^2 - x - 2} = 4$$

So

$$5x^2 - 2x + 1 = 4(x^2 - x - 2)$$

Expanding the right-hand side:

$$5x^2 - 2x + 1 = 4x^2 - 4x - 8$$

Bring all terms to one side:

$$5x^2 - 2x + 1 - 4x^2 + 4x + 8 = 0$$

$$x^2 + 2x + 9 = 0$$

The discriminant is

$$2^2 - 4(1)(9) = 4 - 36 = -32 < 0$$

so the equation has no real solutions.

Therefore there are no real values of x for which $y = 4$, so the line $y = 4$ does not intersect C.

2. A curve C_1 has equation

$$y = -2 + \frac{12}{x+3}$$

(a) Write down the equations of the asymptotes of curve C_1. **[2]**

(b) Sketch the graph of curve C_1. Indicate the values of the intercepts of the curve with the axes. **[3]**

(c) Hence, or otherwise, solve the inequality

$$-2 + \frac{12}{x+3} > 0$$

[2]

(d) Curve C_2 is a reflection of curve C_1 in the line $y = -x$. Find an equation for curve C_2 in the form $y = f(x)$. **[3]**

Solution

(a) The denominator is zero when $x = -3$, so the vertical asymptote is

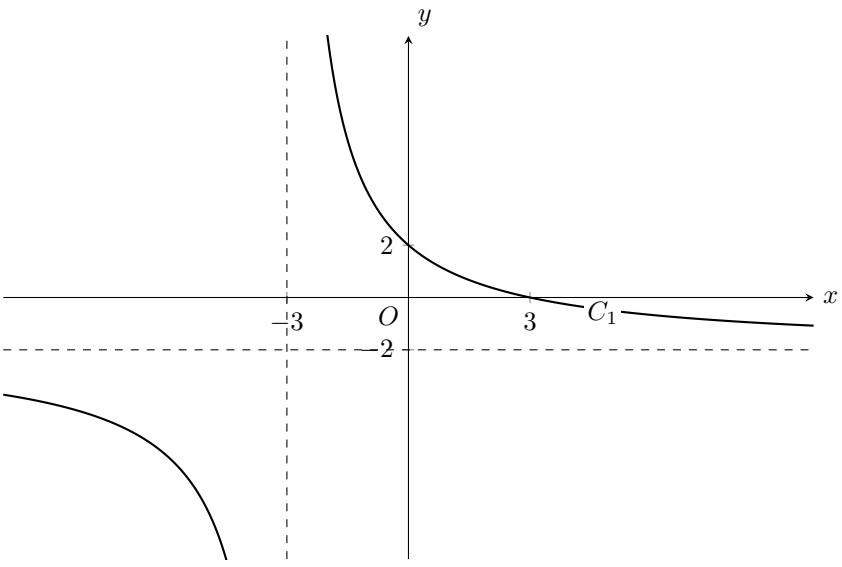
$$x = -3$$

As $x \to \pm\infty$, $\frac{12}{x+3} \to 0$, so the horizontal asymptote is

$$y = -2$$

Therefore the asymptotes are $x = -3$ and $y = -2$.

(b) The sketch should look like:



This is a rectangular hyperbola with centre $(-3, -2)$, with asymptotes $x = -3$ and $y = -2$. For $x > -3$, the branch lies above $y = -2$; for $x < -3$, the branch lies below $y = -2$.

To find the intercepts:

Setting $x = 0$:

$$y = -2 + \frac{12}{0+3} = -2 + 4 = 2$$

so the y-intercept is $(0, 2)$.

Setting $y = 0$:

$$-2 + \frac{12}{x+3} = 0$$

$$-2(x+3) + 12 = 0$$

$$2x = 6$$

$$x = 3$$

so the x-intercept is $(3, 0)$.

Therefore the intercepts are $(3, 0)$ and $(0, 2)$.

(c) We need to solve

$$-2 + \frac{12}{x+3} > 0$$

From the sketch in part (b), this means we want the part of the curve that lies *above* the x-axis.

Looking at the graph, curve C_1 is above the x-axis only between its vertical asymptote at $x = -3$ and its x-intercept at $x = 3$.

Hence

$$-3 < x < 3$$

(d) Reflection in the line $y = -x$ maps each point (x, y) to $(-y, -x)$.

So in the equation of C_1, replace x by $-y$ and y by $-x$:

$$-x = -2 + \frac{12}{-y+3}$$

that is,

$$-x = -2 + \frac{12}{3-y}$$

Now solve for y:

$$-x + 2 = \frac{12}{3-y}$$

$$3 - y = \frac{12}{2-x}$$

Since $2 - x = -(x - 2)$, we have

$$\frac{12}{2-x} = -\frac{12}{x-2}$$

so

$$3 - y = -\frac{12}{x-2}$$

$$y = 3 + \frac{12}{x-2}$$

3. A curve has equation

$$y = \frac{x^2 - 9}{x^2 - 4}$$

(a) Write down the equations of the asymptotes to the curve. **[2]**

(b) Sketch the curve, indicating the coordinates of the points where the curve intersects the coordinate axes. **[5]**

(c) Hence, or otherwise, solve the inequality

$$0 < \frac{x^2 - 9}{x^2 - 4} < 1$$ **[2]**

Solution

(a) The denominator is zero when $x = \pm 2$, so the vertical asymptotes are

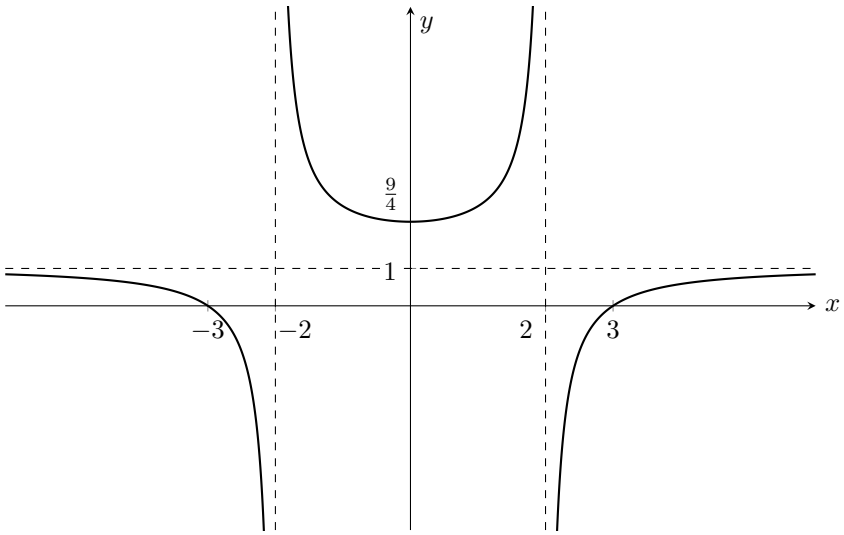
$$x = \pm 2$$

The numerator and denominator have the same degree, so $y \to$ the ratio of the leading coefficients as $x \to \pm\infty$. The horizontal asymptote is

$$y = 1$$

Hence the asymptotes are $x = -2$, $x = 2$ and $y = 1$.

(b) The sketch should look like:



Since the equation depends on x^2, the curve is symmetric about the y-axis.

The x-intercepts are found by setting $y = 0$:

$$\frac{x^2 - 9}{x^2 - 4} = 0 \quad \Longrightarrow \quad x^2 - 9 = 0$$

so

$$x = \pm 3$$

giving intercepts $(-3, 0)$ and $(3, 0)$.

The y-intercept is found by setting $x = 0$:

$$y = \frac{0 - 9}{0 - 4} = \frac{9}{4}$$

so the point is

$$\left(0, \frac{9}{4}\right)$$

(c) Let

$$f(x) = \frac{x^2 - 9}{x^2 - 4}$$

From the sketch in part (b), the inequality

$$0 < f(x) < 1$$

means that the curve must lie *above* the x-axis and *below* the horizontal line $y = 1$.

Looking at the graph, this happens only on the two outer branches, that is for

$$x < -3 \quad \text{or} \quad x > 3$$

4. A curve has equation $y = \dfrac{x-1}{x^2-x-6}$.

(a) (i) Write down the equations of the three asymptotes of the curve. **[2]**

(ii) Sketch the curve, showing the coordinates of any points of intersection with the coordinate axes. **[4]**

(b) Hence, or otherwise, solve the inequality

$$\frac{x-1}{x^2-x-6} > \frac{1}{2} \qquad \textbf{[3]}$$

Solution

(a) (i) Factorising the denominator:

$$x^2 - x - 6 = (x-3)(x+2)$$

The denominator is zero when $x = 3$ or $x = -2$, so the vertical asymptotes are

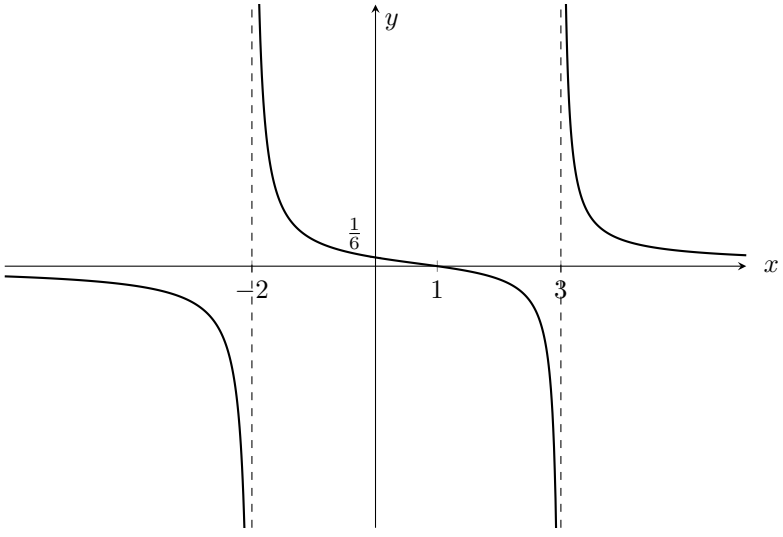
$$x = 3 \quad \text{and} \quad x = -2$$

Since the degree of the denominator is greater than the degree of the numerator, the horizontal asymptote is

$$y = 0$$

Hence the three asymptotes are $x = -2$, $x = 3$ and $y = 0$.

(ii) The sketch should look like:



The intersections with the axes are found as follows.

For the x-axis, $y = 0$, so

$$x - 1 = 0$$

giving $x = 1$. Hence the curve crosses the x-axis at $(1, 0)$.

For the y-axis, $x = 0$, so

$$y = \frac{0-1}{0^2-0-6} = \frac{-1}{-6} = \frac{1}{6}$$

Hence the curve crosses the y-axis at $\left(0, \frac{1}{6}\right)$.

(b) To solve

$$\frac{x-1}{x^2-x-6} > \frac{1}{2}$$

we compare the curve from part (a)(ii) with the line

$$y = \frac{1}{2}$$

First find where the curve meets this line:

$$\frac{x-1}{x^2-x-6} = \frac{1}{2}$$

$$2(x-1) = x^2-x-6$$

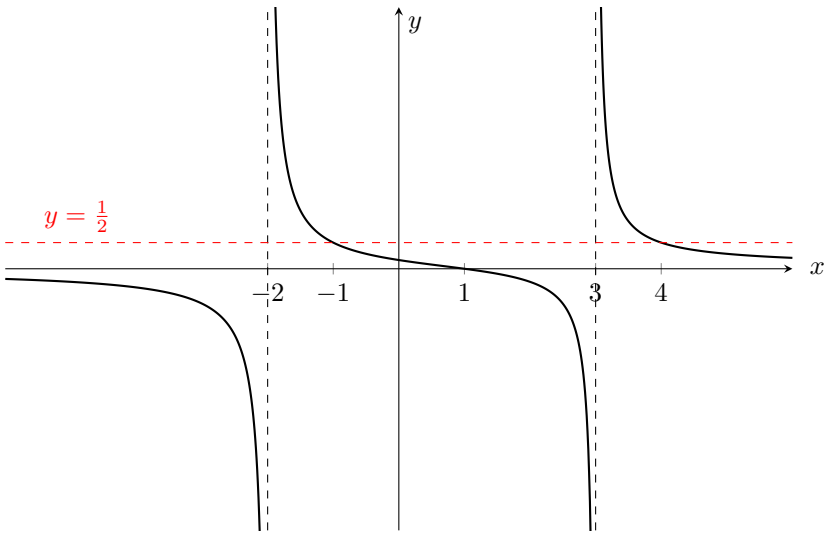
$$x^2-3x-4 = 0$$

$$(x-4)(x+1) = 0$$

So the graphs intersect when

$$x = -1 \quad \text{or} \quad x = 4$$

Adding the line $y = \frac{1}{2}$ to the sketch:



From the graph, the curve lies above the line $y = \frac{1}{2}$ for

$$-2 < x < -1 \quad \text{or} \quad 3 < x < 4$$

5. A curve has equation

$$y = 2 - \frac{3}{x+1}$$

(a) State the equations of the horizontal and vertical asymptotes of this curve. **[2]**

(b) The straight line $y = 3x + c$ meets the curve. Show that the x-coordinates of the points of intersection satisfy

$$3x^2 + (c+1)x + (c+1) = 0$$

[3]

(c) It is given that the line $y = 3x + c$ is a tangent to the curve.

(i) Without using calculus, find the two possible values of c. **[3]**
(ii) Hence find the coordinates of each point of contact. **[4]**

Solution

(a) As $x \to \pm\infty$, $\frac{3}{x+1} \to 0$, so the horizontal asymptote is

$$y = 2$$

The denominator is zero when $x = -1$, so the vertical asymptote is

$$x = -1$$

(b) At a point of intersection, the y-values are equal, so

$$3x + c = 2 - \frac{3}{x+1}$$

Rearranging:

$$3x + c - 2 = -\frac{3}{x+1}$$

Multiplying by $(x+1)$:

$$(3x + c - 2)(x + 1) = -3$$

Expanding:

$$3x^2 + 3x + (c-2)x + (c-2) = -3$$

$$3x^2 + (c+1)x + c - 2 = -3$$

$$3x^2 + (c+1)x + (c+1) = 0$$

Hence the x-coordinates of the points of intersection satisfy

$$3x^2 + (c+1)x + (c+1) = 0$$

(c) (i) If the line is a tangent, the two points of intersection coincide, so the quadratic in part (b) has equal roots.

Therefore its discriminant is zero:

$$(c+1)^2 - 4(3)(c+1) = 0$$

So

$$(c+1)^2 - 12(c+1) = 0$$

$$(c+1)\big((c+1) - 12\big) = 0$$

$$(c+1)(c-11) = 0$$

Hence

$$c = -1 \quad \text{or} \quad c = 11$$

(ii) From part (b), the tangent case gives a repeated root of

$$3x^2 + (c+1)x + (c+1) = 0$$

so

$$x = \frac{-(c+1)}{2 \cdot 3} = -\frac{c+1}{6}$$

If $c = -1$, then

$$x = -\frac{-1+1}{6} = 0$$

Using $y = 3x + c$,

$$y = 3(0) - 1 = -1$$

So one point of contact is

$$(0, -1)$$

If $c = 11$, then

$$x = -\frac{11+1}{6} = -2$$

Using $y = 3x + c$,

$$y = 3(-2) + 11 = 5$$

So the other point of contact is

$$(-2, 5)$$

Hence the points of contact are

$$(0, -1) \quad \text{and} \quad (-2, 5)$$

6. The function f is defined by

$$f(x) = \frac{-2x^2 + 11x - 20}{2x - 3} \qquad (x \in \mathbb{R},\ x \neq \tfrac{3}{2})$$

(a) Find the range of f. **[5]**

(b) Find the coordinates of the two stationary points of the graph of $y = f(x)$. **[3]**

(c) Show that the graph of $y = f(x)$ has an oblique asymptote and find its equation. **[2]**

(d) Sketch the graph of $y = f(x)$, showing clearly the asymptotes and stationary points. **[4]**

Solution

(a) Let $y = f(x)$. Then

$$y = \frac{-2x^2 + 11x - 20}{2x - 3}$$

so

$$y(2x - 3) = -2x^2 + 11x - 20$$

$$2x^2 + (2y - 11)x + (20 - 3y) = 0$$

For a given value of y, this is a quadratic in x. Real values of x exist if and only if the discriminant is non-negative:

$$\begin{aligned}(2y - 11)^2 - 4(2)(20 - 3y) &= 4y^2 - 44y + 121 - 160 + 24y \\ &= 4y^2 - 20y - 39 \\ &= (2y - 13)(2y + 3)\end{aligned}$$

So we require

$$(2y - 13)(2y + 3) \geq 0$$

This gives

$$y \leq -\frac{3}{2} \qquad \text{or} \qquad y \geq \frac{13}{2}$$

We must also check that the solution is not the excluded value $x = \frac{3}{2}$. Substituting $x = \frac{3}{2}$ into

$$2x^2 + (2y - 11)x + (20 - 3y)$$

gives

$$2\left(\frac{3}{2}\right)^2 + (2y - 11)\left(\frac{3}{2}\right) + (20 - 3y) = 8 \neq 0$$

so $x = \frac{3}{2}$ is never a root. Hence every y satisfying the inequality is genuinely taken by the function. Now check the endpoints.

If $y = \frac{13}{2}$, then

$$2x^2 + 2x + \frac{1}{2} = 0$$
$$4x^2 + 4x + 1 = 0$$
$$(2x+1)^2 = 0$$

so $x = -\frac{1}{2}$.

If $y = -\frac{3}{2}$, then

$$2x^2 - 14x + \frac{49}{2} = 0$$
$$4x^2 - 28x + 49 = 0$$
$$(2x-7)^2 = 0$$

so $x = \frac{7}{2}$.

Thus the endpoints are included, so the range of f is given by

$$f(x) \leq -\frac{3}{2} \quad \text{or} \quad f(x) \geq \frac{13}{2}$$

(b) First divide the numerator by the denominator:

$$-2x^2 + 11x - 20 = (2x-3)(-x+4) - 8$$

so

$$f(x) = -x + 4 - \frac{8}{2x-3}$$

Differentiate:

$$\begin{aligned} f'(x) &= \frac{\mathrm{d}}{\mathrm{d}x}\left(-x + 4 - 8(2x-3)^{-1}\right) \\ &= -1 - 8(-1)(2)(2x-3)^{-2} \\ &= -1 + \frac{16}{(2x-3)^2} \end{aligned}$$

For stationary points, set $f'(x) = 0$:

$$-1 + \frac{16}{(2x-3)^2} = 0$$
$$\frac{16}{(2x-3)^2} = 1$$
$$(2x-3)^2 = 16$$

so

$$2x - 3 = \pm 4$$

giving

$$x = -\frac{1}{2} \quad \text{or} \quad x = \frac{7}{2}$$

Now find the corresponding y-values:

$$f\left(-\frac{1}{2}\right) = -\left(-\frac{1}{2}\right) + 4 - \frac{8}{-4} = \frac{1}{2} + 4 + 2 = \frac{13}{2}$$

$$f\left(\frac{7}{2}\right) = -\frac{7}{2} + 4 - \frac{8}{4} = \frac{1}{2} - 2 = -\frac{3}{2}$$

So the stationary points are

$$\left(-\frac{1}{2}, \frac{13}{2}\right) \quad \text{and} \quad \left(\frac{7}{2}, -\frac{3}{2}\right)$$

Alternative Solution:

From part (a), the range boundaries are $y = \frac{13}{2}$ and $y = -\frac{3}{2}$. At either boundary, the quadratic

$$2x^2 + (2y - 11)x + (20 - 3y) = 0$$

has discriminant zero, so its repeated root is

$$x = -\frac{2y - 11}{4} = \frac{11 - 2y}{4}$$

Substituting each boundary value gives

$$y = \frac{13}{2} \quad \Longrightarrow \quad x = \frac{11 - 13}{4} = -\frac{1}{2}$$

$$y = -\frac{3}{2} \quad \Longrightarrow \quad x = \frac{11 + 3}{4} = \frac{7}{2}$$

Thus the points are

$$\left(-\frac{1}{2}, \frac{13}{2}\right) \quad \text{and} \quad \left(\frac{7}{2}, -\frac{3}{2}\right)$$

They attain the two range boundaries, so they are a local minimum and a local maximum, respectively. Since f is differentiable at both points, they are stationary points.

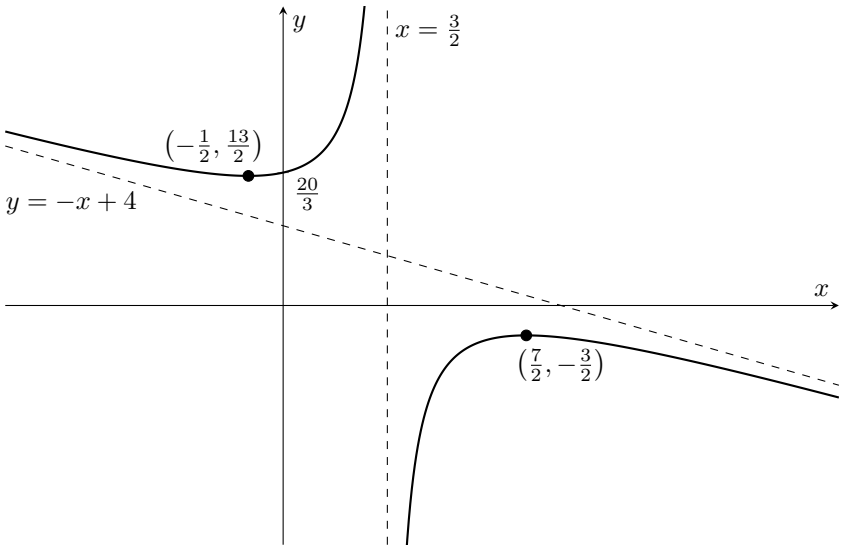
(c) From part (b):

$$f(x) = -x + 4 - \frac{8}{2x - 3}$$

As $x \to \pm\infty$, $\frac{8}{2x-3} \to 0$, so the oblique asymptote is

$$y = -x + 4$$

(d) The sketch should look like:



Key features for the sketch are as follows.

From part (c), the oblique asymptote is

$$y = -x + 4$$

The denominator is zero when $x = \frac{3}{2}$, so the vertical asymptote is

$$x = \frac{3}{2}$$

From part (b), the stationary points are

$$\left(-\frac{1}{2}, \frac{13}{2}\right) \quad \text{and} \quad \left(\frac{7}{2}, -\frac{3}{2}\right)$$

The y-intercept is

$$f(0) = \frac{-20}{-3} = \frac{20}{3}$$

so the graph passes through $\left(0, \frac{20}{3}\right)$.

For x-intercepts, solve

$$-2x^2 + 11x - 20 = 0$$

Its discriminant is

$$11^2 - 4(-2)(-20) = 121 - 160 = -39 < 0$$

so there are no x-intercepts.

7. The function f is defined by

$$f(x) = \frac{x^2 + 2x - 3}{x^2 + px + 4} \qquad x \in \mathbb{R}$$

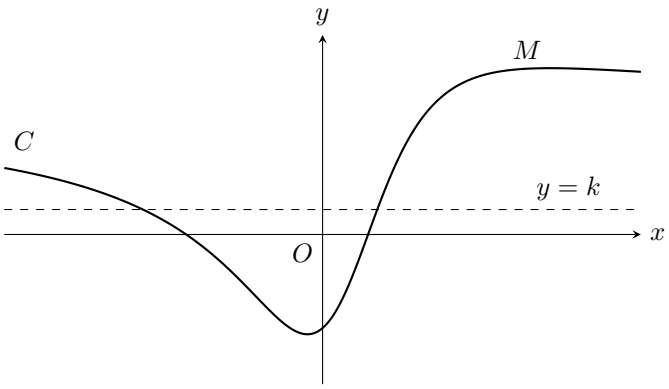
where p is a constant.

The graph of $y = f(x)$ has only one asymptote.

(a) Write down the equation of the asymptote. **[1]**

(b) Find the set of possible values of p. **[4]**

(c) Find the coordinates of the points at which the graph of $y = f(x)$ intersects the coordinate axes. **[3]**



(d) A curve C has equation

$$y = \frac{x^2 + 2x - 3}{x^2 - x + 4}$$

The curve C has a local maximum at the point M as shown in the diagram.
The line $y = k$ intersects curve C.

(i) Show that

$$15k^2 - 8k - 16 \leq 0$$ **[4]**

(ii) Hence, find the y-coordinate of point M. **[2]**

Solution

(a) The numerator and denominator have the same degree, so $y \to$ the ratio of the leading coefficients as $x \to \pm\infty$. The horizontal asymptote is

$$y = 1$$

(b) From part (a), one asymptote is already $y = 1$.

For the graph to have *only one* asymptote, there must be no vertical asymptotes. Vertical asymptotes would occur if the denominator

$$x^2 + px + 4$$

had any real roots.

Therefore we require the quadratic $x^2 + px + 4 = 0$ to have no real solutions, so its discriminant must be negative:

$$\begin{aligned} p^2 - 4(1)(4) &< 0 \\ p^2 - 16 &< 0 \\ p^2 &< 16 \\ -4 < p &< 4 \end{aligned}$$

(c) To find the x-intercepts, set $y = 0$. Then the numerator must be zero:

$$\begin{aligned} x^2 + 2x - 3 &= 0 \\ (x+3)(x-1) &= 0 \\ x = -3 \quad &\text{or} \quad x = 1 \end{aligned}$$

So the possible points are $(-3, 0)$ and $(1, 0)$.

From part (b), the denominator $x^2 + px + 4$ has no real roots, so it is not zero at these values.

Hence the graph intersects the x-axis at

$$(-3, 0) \quad \text{and} \quad (1, 0)$$

To find the y-intercept, set $x = 0$:

$$y = \frac{0^2 + 2(0) - 3}{0^2 + p(0) + 4} = -\frac{3}{4}$$

So the graph intersects the y-axis at

$$\left(0, -\frac{3}{4}\right)$$

(d) (i) The line $y = k$ intersects the curve

$$y = \frac{x^2 + 2x - 3}{x^2 - x + 4}$$

when

$$k = \frac{x^2 + 2x - 3}{x^2 - x + 4}$$

First note that

$$x^2 - x + 4 = \left(x - \frac{1}{2}\right)^2 + \frac{15}{4} > 0$$

for all real x, so we may multiply through safely.

Hence

$$x^2 + 2x - 3 = k(x^2 - x + 4)$$

Rearranging:

$$\begin{aligned} x^2 + 2x - 3 &= kx^2 - kx + 4k \\ 0 &= kx^2 - kx + 4k - x^2 - 2x + 3 \\ 0 &= (k-1)x^2 - (k+2)x + (4k+3) \end{aligned}$$

Equivalently:

$$(1-k)x^2 + (2+k)x - (3+4k) = 0$$

For the line to intersect the curve, this quadratic in x must have at least one real root. So its discriminant is non-negative:

$$(2+k)^2 - 4(1-k)\big(-(3+4k)\big) \geq 0$$

Now expand:

$$\begin{aligned}(2+k)^2 + 4(1-k)(3+4k) &\geq 0\\ (k^2+4k+4) + 4(3+4k-3k-4k^2) &\geq 0\\ k^2+4k+4+4(3+k-4k^2) &\geq 0\\ k^2+4k+4+12+4k-16k^2 &\geq 0\\ -15k^2+8k+16 &\geq 0\end{aligned}$$

Therefore

$$15k^2 - 8k - 16 \leq 0$$

as required.

(ii) At the local maximum, the horizontal line $y = k$ just touches the curve, so there is exactly one value of x. Therefore the discriminant is zero, so

$$15k^2 - 8k - 16 = 0$$

Solving:

$$k = \frac{8 \pm \sqrt{(-8)^2 - 4(15)(-16)}}{2(15)} = \frac{8 \pm \sqrt{64+960}}{30} = \frac{8 \pm 32}{30}$$

So

$$k = \frac{40}{30} = \frac{4}{3} \qquad \text{or} \qquad k = \frac{-24}{30} = -\frac{4}{5}$$

The local maximum has the larger y-value, so the y-coordinate of M is

$$\frac{4}{3}$$

8. The curve C has equation

$$y = \frac{x^2 + 3}{x + 1}$$

(a) Find the equations of the asymptotes of C. **[3]**

(b) Show that there is no point on C for which $-6 < y < 2$. **[5]**

(c) Sketch C. **[2]**

(d) (i) Sketch the graphs of

$$y = \left|\frac{x^2 + 3}{x + 1}\right| \text{ and } y = |x - 1|$$

on the same axes, stating the coordinates of any intersections with the axes. **[4]**

(ii) Use your sketch to find the set of values of c for which

$$\left|\frac{x^2 + 3}{x + 1}\right| \leq |x - 1| + c$$

has no solution. **[1]**

Solution

(a) Divide the numerator by the denominator:

$$x^2 + 3 = (x + 1)(x - 1) + 4$$

so

$$y = \frac{x^2 + 3}{x + 1} = x - 1 + \frac{4}{x + 1}$$

The denominator is zero when $x = -1$, so the vertical asymptote is

$$x = -1$$

As $x \to \pm\infty$, $\frac{4}{x+1} \to 0$, so the oblique asymptote is

$$y = x - 1$$

Therefore the asymptotes are $x = -1$ and $y = x - 1$.

(b) Start from

$$y = \frac{x^2 + 3}{x + 1}$$

and rearrange:

$$y(x + 1) = x^2 + 3$$

$$x^2 - yx + (3 - y) = 0$$

This is a quadratic in x. For real values of x, the discriminant is non-negative:

$$\begin{aligned}(-y)^2 - 4(1)(3-y) &= y^2 - 12 + 4y \\ &= y^2 + 4y - 12 \\ &= (y+6)(y-2)\end{aligned}$$

Hence

$$(y+6)(y-2) \geq 0$$

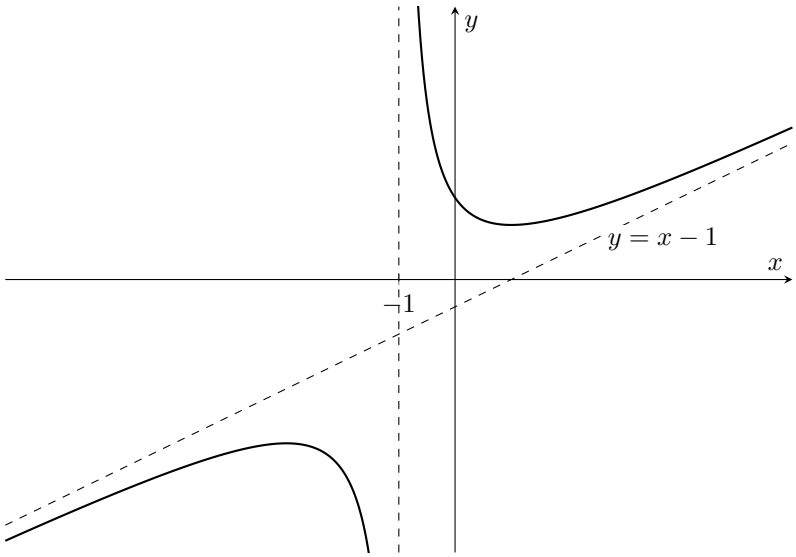
This gives

$$y \leq -6 \quad \text{or} \quad y \geq 2$$

So there is no point on C for which

$$-6 < y < 2$$

(c) The sketch should look like:



Finding the y-intercept:

$$y(0) = 3$$

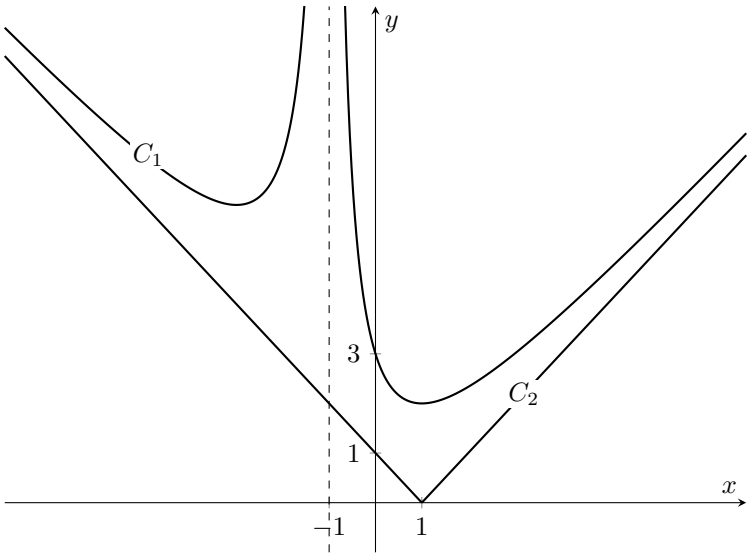
so the y-intercept is $(0, 3)$.

There is no x-intercept because

$$\frac{x^2+3}{x+1} = 0 \quad \Longrightarrow \quad x^2 + 3 = 0$$

which has no real solution.

(d) (i) The two sketches should look like:



Here

$$C_1 : \ y = \left| \frac{x^2+3}{x+1} \right|, \qquad C_2 : \ y = |x-1|$$

For C_1, the part of the original curve with $x < -1$ is reflected in the x-axis. The right-hand branch is unchanged.

Its intersections with the axes are:

$$y = \left| \frac{0^2+3}{0+1} \right| = 3$$

so the y-intercept is $(0, 3)$.

There is no x-intercept, since $x^2 + 3 > 0$ for all real x.

Also, C_1 has vertical asymptote $x = -1$, and its oblique asymptotes are

$$y = 1 - x \quad \text{as } x \to -\infty, \qquad y = x - 1 \quad \text{as } x \to \infty$$

For $C_2 : y = |x - 1|$:

$$|x-1| = 0 \quad \Longrightarrow \quad x = 1$$

so the x-intercept is $(1, 0)$, and

$$y = |0 - 1| = 1$$

so the y-intercept is $(0, 1)$.

(ii) From the sketch, $y = \left| \frac{x^2+3}{x+1} \right|$ lies above $y = |x - 1|$ for all x, and the gap tends to 0 as $x \to \pm\infty$.

For the equality case $c = 0$,

$$\begin{aligned}
\left|\frac{x^2+3}{x+1}\right| = |x-1| \quad &\Longrightarrow \quad \frac{(x^2+3)^2}{(x+1)^2} = (x-1)^2 \\
&\Longrightarrow \quad (x^2+3)^2 = (x-1)^2(x+1)^2 \\
&\Longrightarrow \quad (x^2+3)^2 = (x^2-1)^2 \\
&\Longrightarrow \quad x^4 + 6x^2 + 9 = x^4 - 2x^2 + 1 \\
&\Longrightarrow \quad 8x^2 + 8 = 0
\end{aligned}$$

which is impossible for real x.

So there is no solution when $c = 0$, and therefore no solution for any $c < 0$ either. Since the graphs get arbitrarily close for large positive or negative x, any $c > 0$ gives a solution.

Hence the inequality has no solution for

$$c \leq 0$$

9. The curve C has equation

$$y = \frac{(x-1)(x+3)}{x^2-2x+3}$$

(a) Show that C has no vertical asymptotes and state the equation of the horizontal asymptote of C. **[3]**

(b) Find the coordinates of the stationary points on C. **[4]**

(c) Sketch C, stating the coordinates of the intersections with the axes. **[3]**

(d) Sketch the curve with equation

$$y = \left|\frac{(x-1)(x+3)}{x^2-2x+3}\right|$$

and state the set of values of k for which

$$\left|\frac{(x-1)(x+3)}{x^2-2x+3}\right| = k$$

has 4 distinct real solutions. **[2]**

Solution

(a) The denominator has discriminant

$$(-2)^2 - 4(1)(3) = -8 < 0$$

so it has no real roots. Therefore the denominator is never zero, and C has no vertical asymptotes.

The numerator and denominator have the same degree, so $y \to$ the ratio of the leading coefficients as $x \to \pm\infty$. The horizontal asymptote is

$$y = 1$$

Alternative Solution:

Completing the square gives

$$x^2 - 2x + 3 = (x-1)^2 + 2 > 0$$

for every real x. Thus the denominator is never zero, so C has no vertical asymptotes. The ratio of the leading coefficients again gives the horizontal asymptote $y = 1$.

(b) First expand the numerator:

$$y = \frac{(x-1)(x+3)}{x^2-2x+3} = \frac{x^2+2x-3}{x^2-2x+3}$$

Differentiate using the quotient rule:

$$\frac{\mathrm{d}y}{\mathrm{d}x} = \frac{(2x+2)(x^2-2x+3)-(x^2+2x-3)(2x-2)}{(x^2-2x+3)^2}$$

$$= \frac{(2x^3-2x^2+2x+6)-(2x^3+2x^2-10x+6)}{(x^2-2x+3)^2}$$

$$= \frac{-4x^2+12x}{(x^2-2x+3)^2}$$

$$= \frac{4x(3-x)}{(x^2-2x+3)^2}$$

For stationary points, set $\frac{\mathrm{d}y}{\mathrm{d}x} = 0$.

Since $(x^2-2x+3)^2 > 0$ for all x, this gives

$$4x(3-x) = 0$$

So

$$x = 0 \quad \text{or} \quad x = 3$$

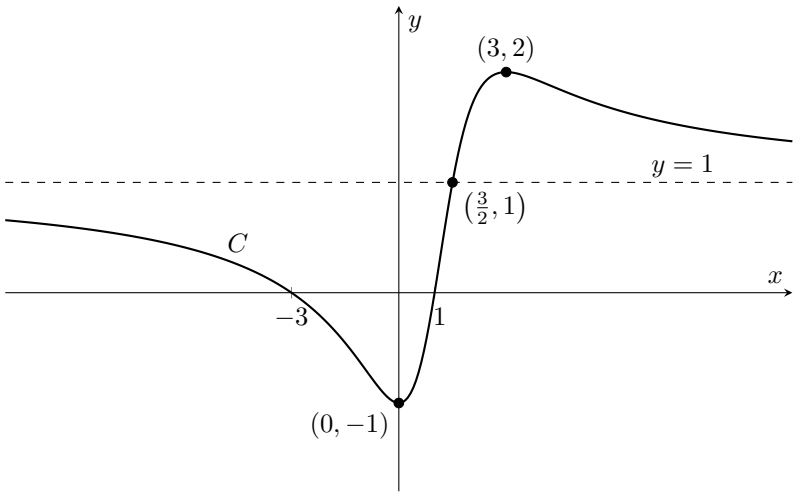
Now find the corresponding y-coordinates:

$$y(0) = \frac{(-1)(3)}{3} = -1, \qquad y(3) = \frac{(2)(6)}{6} = 2$$

So the stationary points are

$$(0,-1) \quad \text{and} \quad (3,2)$$

(c) The sketch should look like:



Intersections with the x-axis occur when $y = 0$, so the numerator must be 0:

$$(x-1)(x+3) = 0$$

Hence

$$x = 1 \quad \text{or} \quad x = -3$$

So the x-intercepts are

$$(1,0) \quad \text{and} \quad (-3,0)$$

The intersection with the y-axis is found by putting $x = 0$:

$$y = \frac{(-1)(3)}{3} = -1$$

So the y-intercept is

$$(0, -1)$$

Using parts (a) and (b), the curve has horizontal asymptote $y = 1$, minimum point $(0, -1)$ and maximum point $(3, 2)$.

It also crosses the horizontal asymptote when $y = 1$:

$$\frac{x^2 + 2x - 3}{x^2 - 2x + 3} = 1$$

$$x^2 + 2x - 3 = x^2 - 2x + 3$$

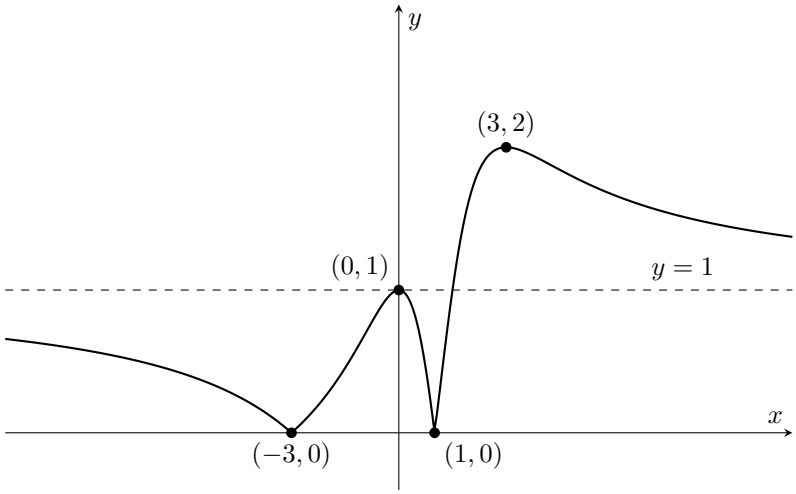
$$4x = 6$$

$$x = \frac{3}{2}$$

So it crosses the asymptote at

$$\left(\frac{3}{2}, 1\right)$$

(d) The sketch should look like:



To solve

$$\left|\frac{(x-1)(x+3)}{x^2 - 2x + 3}\right| = k$$

draw the horizontal line $y = k$.

For there to be 4 distinct real solutions, the line must meet the curve four times: once on the left-hand branch, twice on the middle arch, and once on the right-hand branch.

From the sketch, this happens exactly when the line lies above the x-axis and below the point $(0, 1)$, that is,

$$0 < k < 1$$

10. The curve C has equation $y = \dfrac{x^2 + ax - 6}{x + 1}$, where $a > 0$.

(a) Find the equations of the asymptotes of C. **[3]**

(b) Show that C has no stationary points. **[4]**

(c) Sketch C, stating the coordinates of the point of intersection with the y-axis and labelling the asymptotes. **[3]**

(d) (i) Sketch the curve with equation $y = \left|\dfrac{x^2 + ax - 6}{x + 1}\right|$. **[2]**

(ii) On your sketch in part (i), draw the line $y = a + 1$. **[1]**

(iii) It is given that $\left|\dfrac{x^2 + ax - 6}{x + 1}\right| < a + 1$ for $-11 < x < -3$ and $0 < x < 4$.
Find the value of a. **[2]**

Solution

(a) Divide the numerator by the denominator:

$$x^2 + ax - 6 = (x + 1)(x + a - 1) - (a + 5)$$

so

$$y = \frac{x^2 + ax - 6}{x + 1} = x + a - 1 - \frac{a + 5}{x + 1}$$

The denominator is zero when $x = -1$, so the vertical asymptote is

$$x = -1$$

As $x \to \pm\infty$, $\frac{a+5}{x+1} \to 0$, so the oblique asymptote is

$$y = x + a - 1$$

(b) From part (a):

$$y = x + a - 1 - (a + 5)(x + 1)^{-1}$$

so

$$\frac{\mathrm{d}y}{\mathrm{d}x} = 1 + (a + 5)(x + 1)^{-2} = 1 + \frac{a + 5}{(x + 1)^2}$$

Since $a > 0$, we have $a + 5 > 0$, and for $x \neq -1$,

$$\frac{a + 5}{(x + 1)^2} > 0$$

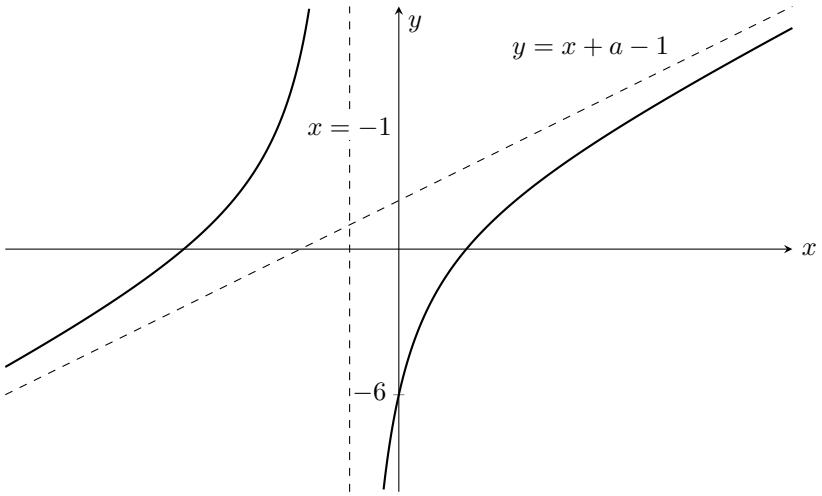
Therefore

$$\frac{\mathrm{d}y}{\mathrm{d}x} > 0 \quad \text{for all } x \neq -1$$

A stationary point would require $\frac{\mathrm{d}y}{\mathrm{d}x} = 0$, which never happens.

Therefore C has no stationary points.

(c) The sketch should look like:

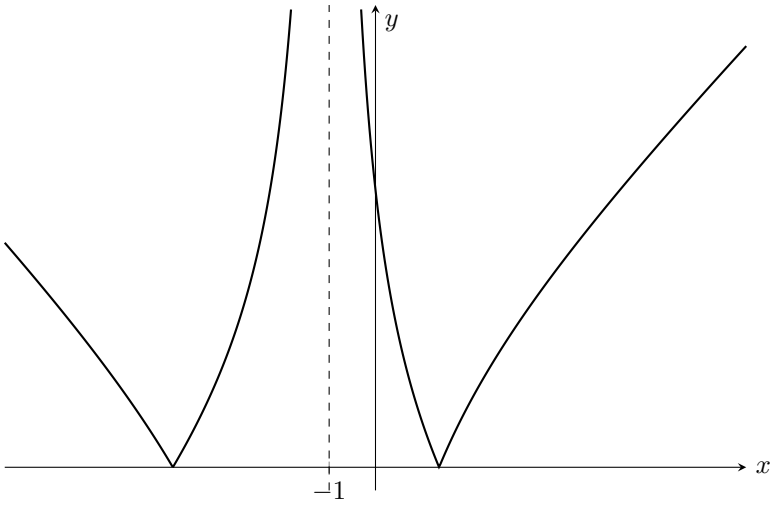


The point where the curve meets the y-axis is found by putting $x = 0$:

$$y = \frac{0^2 + a(0) - 6}{0 + 1} = -6$$

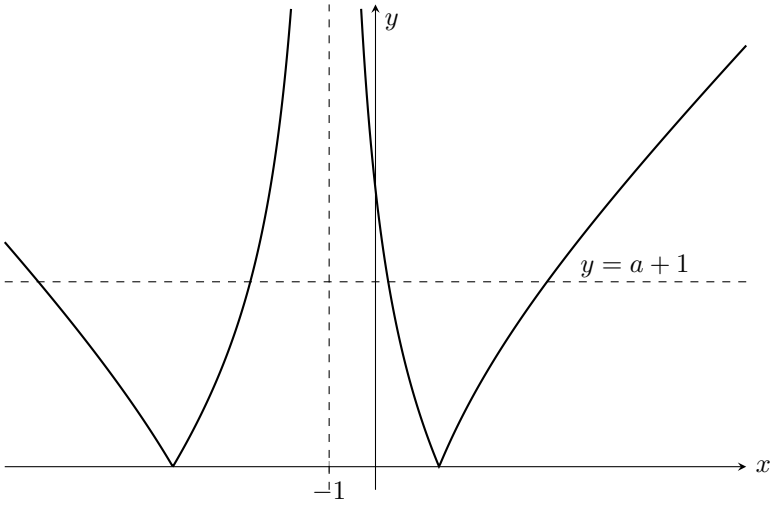
so the point is $(0, -6)$.

(d) (i) The sketch should look like:



This is obtained by reflecting the parts of C below the x-axis in the x-axis.

(ii) Draw the horizontal line $y = a + 1$ on the sketch:



(iii) The endpoints of the given intervals are where

$$\left|\frac{x^2 + ax - 6}{x + 1}\right| = a + 1$$

From the intervals

$$-11 < x < -3 \quad \text{and} \quad 0 < x < 4$$

we can see that $x = 0$ is one boundary point. So

$$\left|\frac{0^2 + a(0) - 6}{0 + 1}\right| = a + 1$$

$$6 = a + 1$$

$$a = 5$$

11. The curve C has equation

$$y = \frac{x^2 - 6x + 5}{x + 2}$$

(a) Find the equations of the asymptotes of C. **[3]**

(b) Find the exact coordinates of the stationary points on C. **[4]**

(c) Sketch C, stating the coordinates of any intersections with the axes. **[3]**

(d) Sketch the curve with equation

$$y = \left| \frac{x^2 - 6x + 5}{x + 2} \right|$$

and find in exact form the set of values of x for which

$$\left| \frac{x^2 - 6x + 5}{x + 2} \right| < 6$$ **[5]**

Solution

(a) Divide the numerator by the denominator:

$$x^2 - 6x + 5 = (x + 2)(x - 8) + 21$$

so

$$y = \frac{x^2 - 6x + 5}{x + 2} = x - 8 + \frac{21}{x + 2}$$

The denominator is zero when $x = -2$, so the vertical asymptote is

$$x = -2$$

As $x \to \pm\infty$, $\frac{21}{x+2} \to 0$, so the oblique asymptote is

$$y = x - 8$$

Hence the asymptotes are $x = -2$ and $y = x - 8$.

(b) Using

$$y = x - 8 + \frac{21}{x + 2}$$

differentiate:

$$\frac{\mathrm{d}y}{\mathrm{d}x} = 1 - 21(x + 2)^{-2} = 1 - \frac{21}{(x + 2)^2}$$

At a stationary point, $\frac{dy}{dx} = 0$, so

$$1 - \frac{21}{(x+2)^2} = 0$$

$$\frac{21}{(x+2)^2} = 1$$

$$(x+2)^2 = 21$$

$$x + 2 = \pm\sqrt{21}$$

Hence

$$x = -2 \pm \sqrt{21}$$

Now find the corresponding y-coordinates from

$$y = x - 8 + \frac{21}{x+2}$$

If $x = -2 - \sqrt{21}$, then $x + 2 = -\sqrt{21}$, so

$$y = (-2 - \sqrt{21}) - 8 + \frac{21}{-\sqrt{21}}$$

$$= -10 - \sqrt{21} - \sqrt{21}$$

$$= -10 - 2\sqrt{21}$$

If $x = -2 + \sqrt{21}$, then $x + 2 = \sqrt{21}$, so

$$y = (-2 + \sqrt{21}) - 8 + \frac{21}{\sqrt{21}}$$

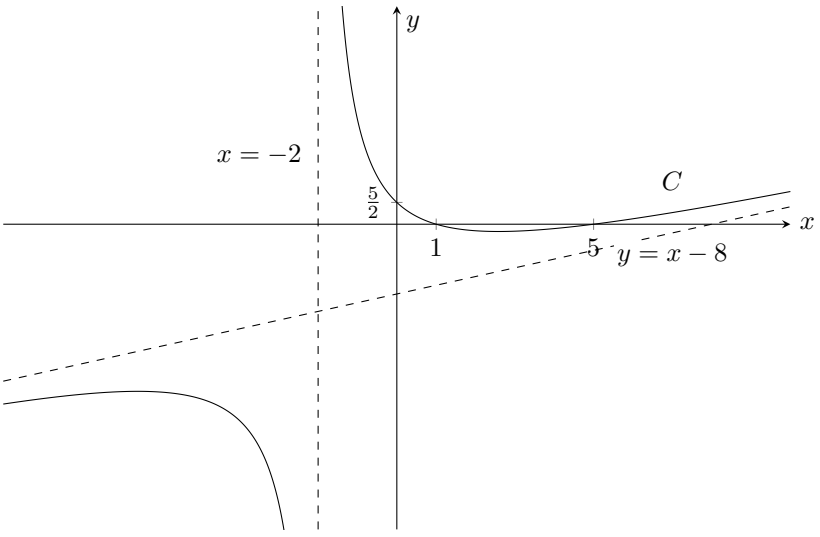
$$= -10 + \sqrt{21} + \sqrt{21}$$

$$= -10 + 2\sqrt{21}$$

So the stationary points are

$$(-2 - \sqrt{21},\ -10 - 2\sqrt{21}) \quad \text{and} \quad (-2 + \sqrt{21},\ -10 + 2\sqrt{21})$$

(c) The sketch should look like:



For the intersections with the axes:

For the x-axis, set $y = 0$:

$$\frac{x^2 - 6x + 5}{x + 2} = 0$$

$$x^2 - 6x + 5 = 0$$

$$(x - 1)(x - 5) = 0$$

So the curve meets the x-axis at

$$(1, 0) \quad \text{and} \quad (5, 0)$$

For the y-axis, set $x = 0$:

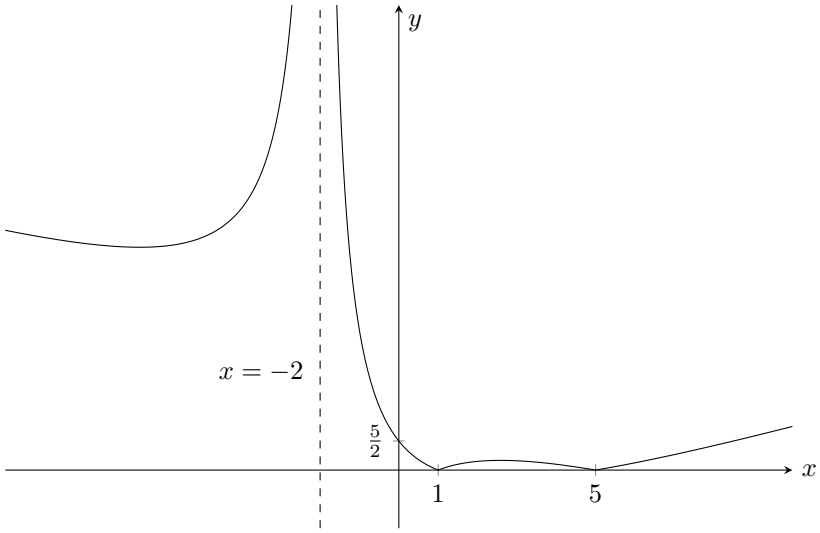
$$y = \frac{0^2 - 6(0) + 5}{0 + 2} = \frac{5}{2}$$

so the curve meets the y-axis at

$$\left(0, \frac{5}{2}\right)$$

The left-hand branch lies below the x-axis for $x < -2$, and the right-hand branch falls from $+\infty$, crosses at $(1, 0)$, reaches the minimum found in part (b), then rises through $(5, 0)$.

(d) The sketch should look like:



For $y = \left|\frac{x^2-6x+5}{x+2}\right|$, every part of the graph of C below the x-axis is reflected in the x-axis. So:

- the whole left-hand branch is reflected above the x-axis
- the part of the right-hand branch between $x = 1$ and $x = 5$ is reflected above the x-axis
- the points $(1, 0)$ and $(5, 0)$ stay on the curve
- $x = -2$ remains a vertical asymptote

Let $f(x) = \frac{x^2-6x+5}{x+2}$. To find where the modulus graph meets $y = 6$, consider $f(x) = \pm 6$.

From part (b), the left branch has maximum $-10 - 2\sqrt{21} < -6$, while the right branch has minimum $-10 + 2\sqrt{21} > -6$. Thus $f(x) = -6$ has no solution.

We therefore solve only $f(x) = 6$:

$$x^2 - 6x + 5 = 6(x + 2)$$

$$x^2 - 12x - 7 = 0$$

$$x = \frac{12 \pm \sqrt{144 + 28}}{2} = 6 \pm \sqrt{43}$$

From the modulus sketch, the curve lies below $y = 6$ between these two intersections. Hence

$$6 - \sqrt{43} < x < 6 + \sqrt{43}$$

Alternative Solution:

We solve

$$\left| \frac{x^2 - 6x + 5}{x + 2} \right| < 6$$

Both sides are non-negative, so we may square:

$$\frac{(x^2 - 6x + 5)^2}{(x + 2)^2} < 36$$

For $x \neq -2$, $(x + 2)^2 > 0$, so multiplying through gives

$$(x^2 - 6x + 5)^2 < 36(x + 2)^2$$

Rearranging as a difference of two squares:

$$(x^2 - 6x + 5)^2 - 36(x + 2)^2 < 0$$

$$\big((x^2 - 6x + 5) - 6(x + 2)\big)\big((x^2 - 6x + 5) + 6(x + 2)\big) < 0$$

$$(x^2 - 12x - 7)(x^2 + 17) < 0$$

But $x^2 + 17 > 0$ for all real x, so we need

$$x^2 - 12x - 7 < 0$$

Solve $x^2 - 12x - 7 = 0$:

$$x = \frac{12 \pm \sqrt{(-12)^2 - 4(1)(-7)}}{2}$$

$$= \frac{12 \pm \sqrt{144 + 28}}{2}$$

$$= \frac{12 \pm \sqrt{172}}{2}$$

$$= 6 \pm \sqrt{43}$$

Since the quadratic opens upwards, it is negative between its roots. Therefore

$$6 - \sqrt{43} < x < 6 + \sqrt{43}$$

12. The curve C has equation

$$y = \frac{-x^2 + 4x + 5}{x + 2}$$

(a) Find the equations of the asymptotes of C. **[3]**

(b) Find the coordinates of the stationary points of C. **[4]**

(c) Sketch C, stating the coordinates of the intersections with the axes. **[3]**

(d) Sketch the curve with equation

$$y = \left|\frac{-x^2 + 4x + 5}{x + 2}\right| \quad \textbf{[2]}$$

(e) Using your sketch, or otherwise, find the set of values of x for which

$$\left|\frac{-x^2 + 4x + 5}{x + 2}\right| < 2 \quad \textbf{[4]}$$

Solution

(a) Divide the numerator by the denominator:

$$-x^2 + 4x + 5 = (x + 2)(-x + 6) - 7$$

so

$$y = -x + 6 - \frac{7}{x + 2}$$

The denominator is zero when $x = -2$, so the vertical asymptote is

$$x = -2$$

As $x \to \pm\infty$, $\frac{7}{x+2} \to 0$, so the oblique asymptote is

$$y = -x + 6$$

Therefore the asymptotes are $x = -2$ and $y = -x + 6$.

(b) Using

$$y = -x + 6 - \frac{7}{x + 2}$$

differentiate:

$$\frac{\mathrm{d}y}{\mathrm{d}x} = -1 + \frac{7}{(x + 2)^2}$$

At a stationary point,

$$-1 + \frac{7}{(x + 2)^2} = 0$$

so

$$\frac{7}{(x+2)^2} = 1$$
$$(x+2)^2 = 7$$
$$x+2 = \pm\sqrt{7}$$
$$x = -2 \pm \sqrt{7}$$

Now find the corresponding y-values.

For $x = -2 - \sqrt{7}$:

$$y = -(-2-\sqrt{7}) + 6 - \frac{7}{-\sqrt{7}}$$
$$= 2 + \sqrt{7} + 6 + \sqrt{7}$$
$$= 8 + 2\sqrt{7}$$

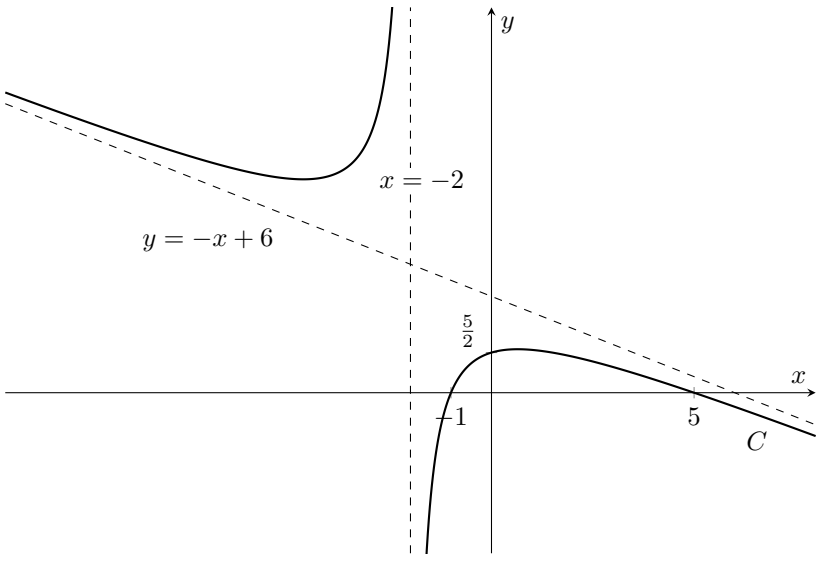
For $x = -2 + \sqrt{7}$:

$$y = -(-2+\sqrt{7}) + 6 - \frac{7}{\sqrt{7}}$$
$$= 2 - \sqrt{7} + 6 - \sqrt{7}$$
$$= 8 - 2\sqrt{7}$$

Therefore the stationary points are

$$(-2-\sqrt{7},\ 8+2\sqrt{7}) \quad \text{and} \quad (-2+\sqrt{7},\ 8-2\sqrt{7})$$

(c) Using the asymptotes and stationary points from earlier parts, the sketch should look like:



The intersections with the x-axis satisfy $y = 0$, so

$$\frac{-x^2+4x+5}{x+2} = 0$$

hence

$$-x^2 + 4x + 5 = 0$$

and

$$x^2 - 4x - 5 = 0$$

$$(x-5)(x+1) = 0$$

so the x-intercepts are

$$(-1, 0) \quad \text{and} \quad (5, 0)$$

The intersection with the y-axis is found by putting $x = 0$:

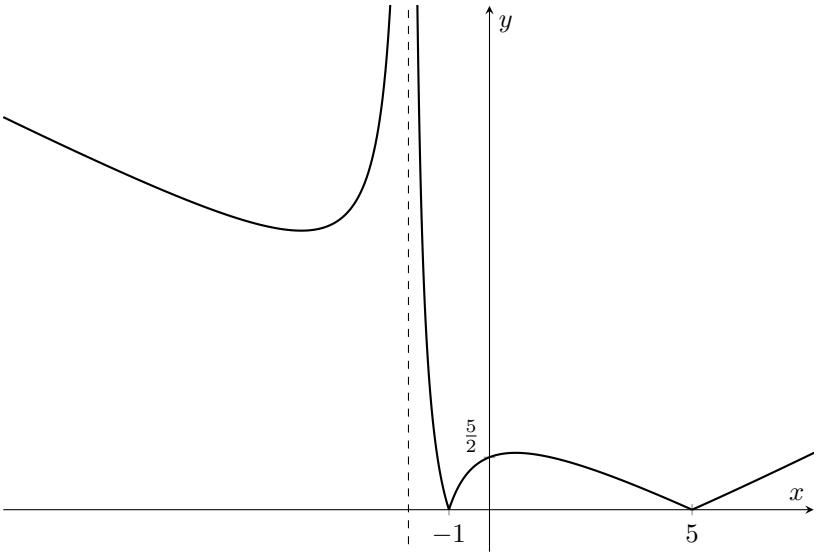
$$y = \frac{5}{2}$$

so the y-intercept is

$$\left(0, \frac{5}{2}\right)$$

Therefore the axis intersections are $(-1, 0)$, $(5, 0)$ and $\left(0, \frac{5}{2}\right)$.

(d) The sketch should look like:



This graph is obtained by reflecting the parts of C below the x-axis in the x-axis.

From the sketch of C, those parts are for

$$-2 < x < -1 \qquad \text{or} \qquad x > 5$$

The zeros stay at $(-1, 0)$ and $(5, 0)$, the vertical asymptote remains $x = -2$, and for large positive x the reflected branch has asymptote

$$y = x - 6$$

(e) Using the sketch in part (d), we need the graph of

$$y = \left| \frac{-x^2 + 4x + 5}{x + 2} \right|$$

to lie below the line $y = 2$.

First find where it meets $y = 2$, so solve

$$\left|\frac{-x^2+4x+5}{x+2}\right| = 2$$

This means solving

$$\frac{-x^2+4x+5}{x+2} = 2 \qquad \text{and} \qquad \frac{-x^2+4x+5}{x+2} = -2$$

For $\frac{-x^2+4x+5}{x+2} = 2$:

$$-x^2+4x+5 = 2x+4$$

$$-x^2+2x+1 = 0$$

$$x^2-2x-1 = 0$$

so

$$x = 1 \pm \sqrt{2}$$

For $\frac{-x^2+4x+5}{x+2} = -2$:

$$-x^2+4x+5 = -2x-4$$

$$-x^2+6x+9 = 0$$

$$x^2-6x-9 = 0$$

so

$$x = 3 \pm 3\sqrt{2}$$

In order, these values are

$$3-3\sqrt{2}, \quad 1-\sqrt{2}, \quad 1+\sqrt{2}, \quad 3+3\sqrt{2}$$

From the sketch, $\left|\frac{-x^2+4x+5}{x+2}\right| < 2$ between the first two intersection points, and between the last two intersection points:

$$3-3\sqrt{2} < x < 1-\sqrt{2} \qquad \text{or} \qquad 1+\sqrt{2} < x < 3+3\sqrt{2}$$

13. The curve C has equation $y = \dfrac{x+1}{x^2-4x+3}$.

(a) Find the equations of the asymptotes of C. **[2]**

(b) Find the coordinates of any stationary points on C. **[4]**

(c) Sketch C, stating the coordinates of the intersections with the axes. **[3]**

(d) Sketch the curve with equation $y = \left|\dfrac{x+1}{x^2-4x+3}\right|$ and find in exact form the set of values of x for which $2|x+1| > |x^2-4x+3|$. **[5]**

Solution

(a) Factorising the denominator:

$$x^2-4x+3 = (x-1)(x-3)$$

The denominator is zero when $x = 1$ or $x = 3$, so the vertical asymptotes are

$$x = 1 \quad \text{and} \quad x = 3$$

Since the degree of the denominator is greater than the degree of the numerator, $y \to 0$ as $x \to \pm\infty$. Hence the horizontal asymptote is

$$y = 0$$

So the asymptotes are $x = 1$, $x = 3$ and $y = 0$.

(b) Using the quotient rule:

$$\begin{aligned}\frac{\mathrm{d}y}{\mathrm{d}x} &= \frac{(x^2-4x+3)(1)-(x+1)(2x-4)}{(x^2-4x+3)^2} \\ &= \frac{x^2-4x+3-(2x^2-2x-4)}{(x^2-4x+3)^2} \\ &= \frac{7-2x-x^2}{(x^2-4x+3)^2}\end{aligned}$$

Stationary points occur when $\frac{\mathrm{d}y}{\mathrm{d}x} = 0$, so

$$7-2x-x^2 = 0$$

$$x^2+2x-7 = 0$$

$$x = \frac{-2 \pm \sqrt{4+28}}{2} = -1 \pm 2\sqrt{2}$$

To find the corresponding y-values, use

$$x^2+2x-7 = 0 \quad \Longrightarrow \quad x^2 = 7-2x$$

so

$$x^2-4x+3 = (7-2x)-4x+3 = 10-6x$$

Hence

$$y = \frac{x+1}{10-6x}$$

For $x = -1 - 2\sqrt{2}$:

$$\begin{aligned} y &= \frac{-2\sqrt{2}}{10-6(-1-2\sqrt{2})} \\ &= \frac{-2\sqrt{2}}{16+12\sqrt{2}} \\ &= \frac{-2\sqrt{2}(16-12\sqrt{2})}{16^2-(12\sqrt{2})^2} \\ &= \frac{-32\sqrt{2}+48}{-32} \\ &= \sqrt{2}-\frac{3}{2} \end{aligned}$$

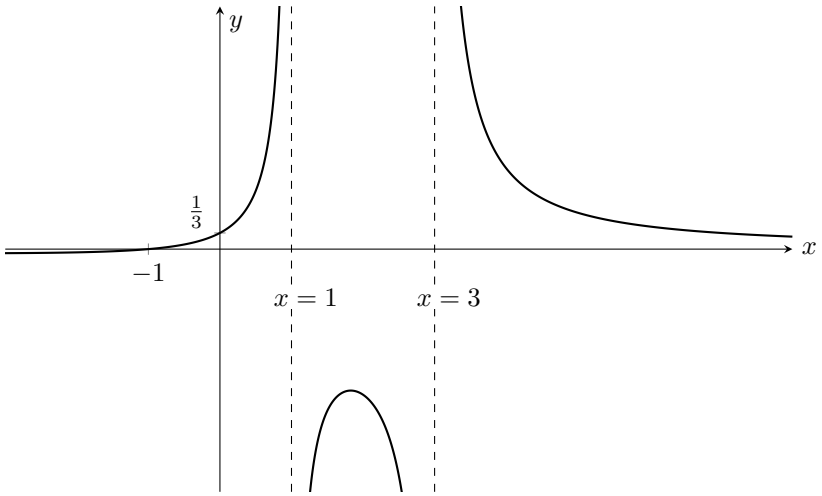
For $x = -1 + 2\sqrt{2}$:

$$\begin{aligned} y &= \frac{2\sqrt{2}}{10-6(-1+2\sqrt{2})} \\ &= \frac{2\sqrt{2}}{16-12\sqrt{2}} \\ &= \frac{2\sqrt{2}(16+12\sqrt{2})}{16^2-(12\sqrt{2})^2} \\ &= \frac{32\sqrt{2}+48}{-32} \\ &= -\sqrt{2}-\frac{3}{2} \end{aligned}$$

Therefore the stationary points are

$$\left(-1-2\sqrt{2},\ \sqrt{2}-\frac{3}{2}\right) \quad \text{and} \quad \left(-1+2\sqrt{2},\ -\sqrt{2}-\frac{3}{2}\right)$$

(c) The sketch should look like:



For the x-axis, set $y = 0$. Then

$$\frac{x+1}{x^2-4x+3} = 0 \quad \Longrightarrow \quad x+1=0 \quad \Longrightarrow \quad x=-1$$

so the x-intercept is

$$(-1, 0)$$

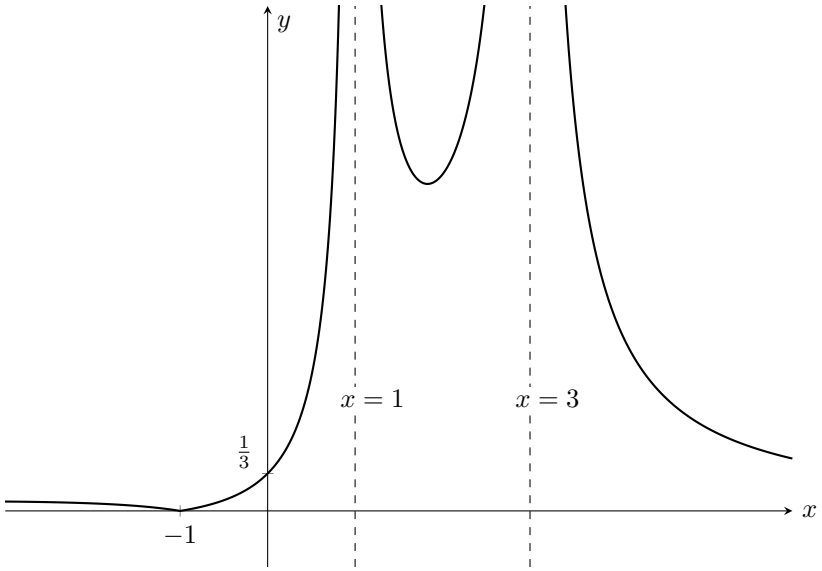
For the y-axis, set $x = 0$. Then

$$y = \frac{0+1}{0-0+3} = \frac{1}{3}$$

so the y-intercept is

$$\left(0, \frac{1}{3}\right)$$

(d) The sketch should look like:



This is the graph from part (c) with the parts below the x-axis reflected in the x-axis. So it has the same asymptotes $x = 1$, $x = 3$, $y = 0$, and there is a cusp at $(-1, 0)$.

Write $f(x) = \frac{x+1}{x^2-4x+3}$. For $x \neq 1, 3$, the inequality is equivalent to

$$|f(x)| > \frac{1}{2}$$

First find where the modulus graph meets $y = \frac{1}{2}$, by solving $f(x) = \pm\frac{1}{2}$:

$$\begin{aligned} f(x) = \frac{1}{2} \quad &\Longrightarrow \quad 2(x+1) = x^2 - 4x + 3 \\ &\Longrightarrow \quad x^2 - 6x + 1 = 0 \\ &\Longrightarrow \quad x = 3 \pm 2\sqrt{2} \end{aligned}$$

The other equation gives

$$\begin{aligned} f(x) = -\frac{1}{2} \quad &\Longrightarrow \quad 2(x+1) = -(x^2 - 4x + 3) \\ &\Longrightarrow \quad x^2 - 2x + 5 = 0 \end{aligned}$$

which has no real roots, since $x^2 - 2x + 5 = (x-1)^2 + 4 > 0$.

From the sketch, $|f(x)| > \frac{1}{2}$ between the two crossings, apart from the undefined points $x = 1, 3$.

At $x = 1$ and $x = 3$, the original inequality has right-hand side zero and left-hand side positive, so both values are included. Therefore

$$3 - 2\sqrt{2} < x < 3 + 2\sqrt{2}$$

Alternative Solution:

Both sides of

$$2|x+1| > |x^2 - 4x + 3|$$

are non-negative, so squaring keeps the inequality in the same direction:

$$\begin{aligned} 4(x+1)^2 &> (x^2 - 4x + 3)^2 \\ 0 &> (x^2 - 4x + 3)^2 - 4(x+1)^2 \\ &= \big(x^2 - 4x + 3 - 2(x+1)\big)\big(x^2 - 4x + 3 + 2(x+1)\big) \\ &= (x^2 - 6x + 1)(x^2 - 2x + 5) \end{aligned}$$

Now

$$x^2 - 2x + 5 = (x-1)^2 + 4 > 0 \quad \text{for all } x$$

so we need

$$x^2 - 6x + 1 < 0$$

Solving $x^2 - 6x + 1 = 0$:

$$x = \frac{6 \pm \sqrt{36 - 4}}{2} = 3 \pm 2\sqrt{2}$$

Since the quadratic opens upwards, it is negative between its roots. Therefore

$$3 - 2\sqrt{2} < x < 3 + 2\sqrt{2}$$

14. Let a be a positive constant.

(a) The curve C_1 has equation

$$y = \frac{x^2 - a^2}{x^2 - 4a^2}$$

Sketch C_1. **[2]**

The curve C_2 has equation

$$y = \left(\frac{x^2 - a^2}{x^2 - 4a^2}\right)^2$$

The curve C_3 has equation

$$y = \left|\frac{x^2 - a^2}{x^2 - 4a^2}\right|$$

(b) (i) Find the coordinates of all stationary points of C_2. **[4]**

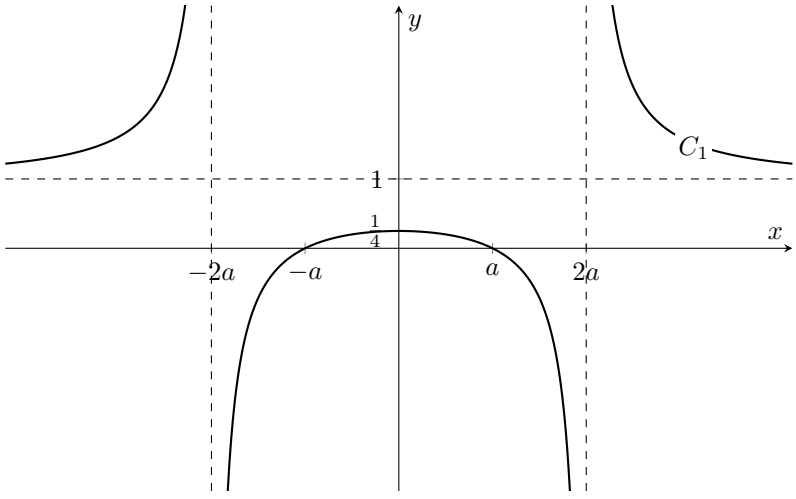
(ii) Find also the coordinates of all points of intersection of C_2 and C_3. **[3]**

(c) Sketch C_2 and C_3 on a single diagram, clearly identifying each curve. Hence find the set of values of x for which

$$\left(\frac{x^2 - a^2}{x^2 - 4a^2}\right)^2 \leq \left|\frac{x^2 - a^2}{x^2 - 4a^2}\right| \quad \textbf{[5]}$$

Solution

(a) The sketch should look like:



Since the equation involves only x^2, C_1 is even, so it is symmetric about the y-axis.

The denominator is zero when $x = \pm 2a$, so the vertical asymptotes are

$$x = \pm 2a$$

The numerator and denominator have the same degree, so $y \to$ the ratio of the leading coefficients as $x \to \pm\infty$. The horizontal asymptote is

$$y = 1$$

The intercepts are:

$$y = 0 \quad \Longrightarrow \quad x^2 - a^2 = 0 \quad \Longrightarrow \quad x = \pm a$$

so $(\pm a, 0)$, and

$$x = 0 \quad \Longrightarrow \quad y = \frac{-a^2}{-4a^2} = \frac{1}{4}$$

so $(0, \frac{1}{4})$.

(b) (i) Let

$$g(x) = \left(\frac{x^2 - a^2}{x^2 - 4a^2}\right)^2$$

Write

$$f(x) = \frac{x^2 - a^2}{x^2 - 4a^2}$$

so that $g(x) = \big(f(x)\big)^2$.

Then

$$g'(x) = 2f(x)f'(x)$$

and, using the quotient rule:

$$\begin{aligned} f'(x) &= \frac{(2x)(x^2 - 4a^2) - (x^2 - a^2)(2x)}{(x^2 - 4a^2)^2} \\ &= \frac{2x\big((x^2 - 4a^2) - (x^2 - a^2)\big)}{(x^2 - 4a^2)^2} \\ &= \frac{2x(-3a^2)}{(x^2 - 4a^2)^2} \\ &= -\frac{6a^2x}{(x^2 - 4a^2)^2} \end{aligned}$$

So

$$\begin{aligned} g'(x) &= 2\left(\frac{x^2 - a^2}{x^2 - 4a^2}\right)\left(-\frac{6a^2x}{(x^2 - 4a^2)^2}\right) \\ &= -\frac{12a^2x(x^2 - a^2)}{(x^2 - 4a^2)^3} \end{aligned}$$

Stationary points occur when $g'(x) = 0$, so

$$-12a^2x(x^2 - a^2) = 0$$

Hence

$$x = 0 \quad \text{or} \quad x^2 = a^2 \quad \Longrightarrow \quad x = \pm a$$

Now find the corresponding y-coordinates:

$$g(0) = \left(\frac{-a^2}{-4a^2}\right)^2 = \left(\frac{1}{4}\right)^2 = \frac{1}{16}$$

and

$$g(\pm a) = 0$$

Therefore the stationary points are

$$\left(0, \frac{1}{16}\right), \quad (-a, 0), \quad (a, 0)$$

(ii) The curves C_2 and C_3 have equations

$$y = \left(\frac{x^2 - a^2}{x^2 - 4a^2}\right)^2 \qquad \text{and} \qquad y = \left|\frac{x^2 - a^2}{x^2 - 4a^2}\right|$$

Let

$$u = \left|\frac{x^2 - a^2}{x^2 - 4a^2}\right|$$

Then intersections satisfy

$$u^2 = u$$

so

$$u(u - 1) = 0$$

Hence

$$u = 0 \quad \text{or} \quad u = 1$$

If $u = 0$, then

$$\left|\frac{x^2 - a^2}{x^2 - 4a^2}\right| = 0$$

so

$$x^2 - a^2 = 0 \quad \Longrightarrow \quad x = \pm a$$

and then $y = 0$.

So two intersection points are

$$(\pm a, 0)$$

If $u = 1$, then

$$\left|\frac{x^2 - a^2}{x^2 - 4a^2}\right| = 1$$

so

$$\frac{x^2 - a^2}{x^2 - 4a^2} = \pm 1$$

First, if

$$\frac{x^2 - a^2}{x^2 - 4a^2} = 1$$

then

$$x^2 - a^2 = x^2 - 4a^2$$

which gives

$$-a^2 = -4a^2$$

impossible.

Next, if

$$\frac{x^2 - a^2}{x^2 - 4a^2} = -1$$

then

$$\begin{aligned} x^2 - a^2 &= -(x^2 - 4a^2) \\ x^2 - a^2 &= -x^2 + 4a^2 \\ 2x^2 &= 5a^2 \\ x^2 &= \frac{5a^2}{2} \end{aligned}$$

so

$$x = \pm a\sqrt{\frac{5}{2}}$$

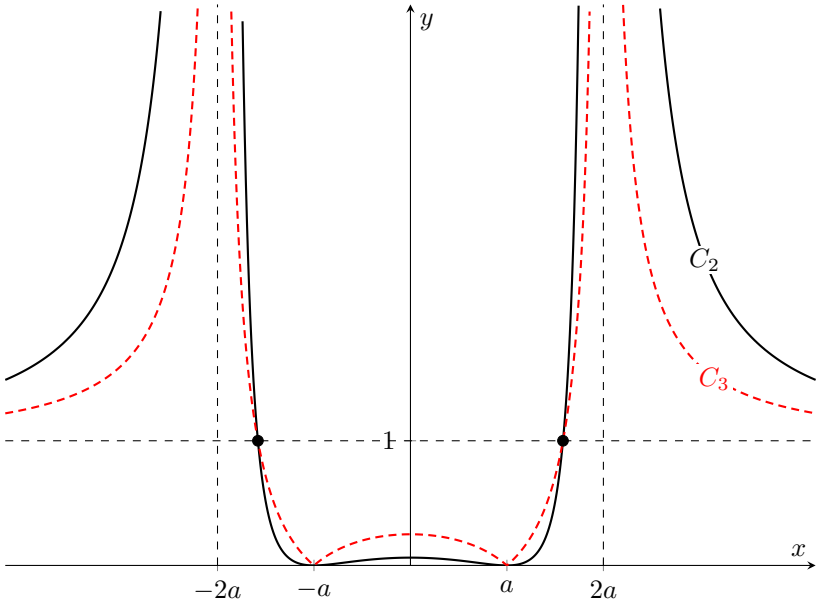
and since $u = 1$,

$$y = 1$$

Therefore the points of intersection are

$$(\pm a, 0), \quad \left(\pm a\sqrt{\frac{5}{2}}, 1\right)$$

(c) The sketch should look like:



Both curves are even, so both are symmetric about the y-axis.

They have the same vertical asymptotes $x = \pm 2a$ and the same horizontal asymptote $y = 1$.

From part (b)(i), C_2 has stationary points

$$\left(0, \frac{1}{16}\right), \quad (\pm a, 0)$$

and from part (b)(ii), the curves meet at

$$(\pm a, 0), \quad \left(\pm a\sqrt{\frac{5}{2}}, 1\right)$$

Also, C_3 is C_1 with the parts below the x-axis reflected in the x-axis, so it passes through $(0, \frac{1}{4})$ and has cusps at $(\pm a, 0)$.

From the sketch, C_2 lies below or on C_3 between the outer intersection points. Therefore

$$\left(\frac{x^2 - a^2}{x^2 - 4a^2}\right)^2 \leq \left|\frac{x^2 - a^2}{x^2 - 4a^2}\right|$$

for

$$-a\sqrt{\frac{5}{2}} \leq x \leq a\sqrt{\frac{5}{2}}$$